Guidance for Assessment Part II

SOST20131 & SOST30031

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1 Introduction

This document provides general guidance for Assessment Part II. It is designed to support you in structuring your report and in explaining how you carried out your analysis, without giving you the answers.

You must:

- Answer both Question 1 (multiple regression) and Question 2 (logistic regression).
- Produce one report covering both questions, up to 2,000 words in total.
- Justify your modelling choices using appropriate graphs, tests, and reasoning.
- Refer to lecture material where relevant (you do not need to formally cite lecture notes).

The overall aim is not only to get "the right model", but to show that you understand the modelling process and can communicate it clearly.

1.1 How to Structure Your Report

A good general structure (for each question) is:

```
1.  Research question(s)
2.  Data description and exploratory analysis
3.  Initial model specification
4.  Model selection and refinement
5.  Final model presentation and interpretation
6.  Prediction and/or model accuracy
7.  Limitations and brief conclusion
```

You can mirror this structure with clear headings and subheadings in your report (they do not need to be identical to the ones below, but the logic should be similar).

2 Question 1: Multiple Regression Guidance

2.1 1. Research Question(s)

Clearly state what you are trying to understand or predict. For example:

- What factors are associated with higher Sales Performance scores?
- How do test scores and background characteristics relate to Sales Performance?

You should express this both in plain language and in statistical language (response variable (Y), explanatory variables ($X_1, X_2, \dots$)).

2.2 2. Data Description and Exploratory Analysis

Describe the dataset and variables:

- Identify the response variable (Sales_Perf).
- Identify each explanatory variable and whether it is:
  - Measured (e.g. test scores),
  - Categorical / attribute (e.g. Type, Level, Expertise).

Suggested steps:

- Show basic summary statistics (mean, SD, min, max) for measured variables.
- Comment on the distribution (e.g. symmetric / skewed, outliers).
- Use appropriate plots, for example:
  - Histogram or boxplot of Sales_Perf.
  - Scatterplots of Sales_Perf vs each measured explanatory variable.
  - Side-by-side boxplots of Sales_Perf by categorical variables.
- Comment on the bivariate relationships:
  - Direction: positive / negative.
  - Strength: strong / moderate / weak (based on plots and correlations).
  - Check for correlations between explanatory variables (for multicollinearity).

You should relate the initial findings to common sense or prior expectations where possible (e.g. "We might expect higher maths scores to be associated with higher sales performance; the scatterplot appears to support this").

2.3 3. Initial Model Specification

Set up a full multiple regression model with Sales_Perf as the response:

$$Sales_Perf = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k + \varepsilon$$

Key points:

- Explain how categorical variables enter the model using dummy variables.
- State clearly which category is the reference group.
- Briefly justify including all variables initially (e.g. theory, instructions in the question, or exploratory evidence).

You do not need to show R output in detail, but you should:

- Report the main results from summary(`lm(...)`):
  - Estimates of coefficients (at least rounded).
  - Standard errors, `t-values`, `p-values`.
  - (R^2) and adjusted (R^2).
    - `F-statistic` and its `p-value`.

2.4 4. Model Selection and Refinement

Explain how you move from the full model to a more parsimonious (simpler but still effective) model.

Consider:

- Overall model fit:
  - Is the F-test significant?
- Does the model explain a reasonable proportion of variability in Sales_Perf (using (R^2) and adjusted (R^2))?
- Individual coefficients:
  - Are some variables clearly non-significant (large p-values)?
  - Do any coefficients have signs that are unexpected or inconsistent with other evidence?
- Multicollinearity:
  - Are there high correlations among explanatory variables?
  - Might multicollinearity be affecting estimates?

Typical approach:

- Remove one non-significant variable at a time and refit.
- After each step:
  - Comment on changes in adjusted (R^2).
  - Check whether remaining variables are still meaningful.
- Stop when you have a model where:
  - Variables are jointly significant.
  - Coefficients are individually interpretable.
  - Adjusted (R^2) is reasonable.
  - There are no obvious multicollinearity issues.

Explain your decisions in words, not just by quoting numbers.

2.5 5. Final Model and Interpretation

Present your final chosen regression model, for example:

$$\widehat{\text{Sales_Perf}} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \cdots + \hat{\beta}_k X_k$$

and explain:

- What each coefficient means (holding other variables constant).
- For dummy variables, interpret relative to the reference category.
- Whether the signs and magnitudes are sensible.

Also comment briefly on model diagnostics, such as:

- Residual plots (any obvious pattern?).
- Any serious outliers or influential observations (only if you checked).

2.6 6. Prediction and Comments on Quality

Use the final model to produce the requested prediction for a given employee profile (the question provides specific values).

You should:

- Plug the given values into the fitted equation to obtain predicted Sales_Perf.
- Show at least the structure of the calculation in the report (you can present the final numeric result rounded).
- Comment on how reliable or uncertain you think this prediction is, using:
  - The overall fit (e.g. adjusted (R^2)).
  - The sample size.
  - Whether the given profile is "typical" of the data or more extreme.
  - Any large residual variability.

3 Question 2: Logistic Regression Guidance

3.1 1. Research Question(s)

Clarify what you are modelling:

- Aim: to model the probability that an individual has diabetes / prediabetes based on a set of health indicators.
- Response variable: Diabetes (0 = no diabetes, 1 = prediabetes or diabetes).
- Explanatory variables: HighBP, HighChol, BMI, Smoker, Fruits, Age.

Express this both in words and using notation (probability (p = P(Diabetes=1))).

3.2 2. Data Description and Exploration

Describe the variables:

- Binary indicators (HighBP, HighChol, Smoker, Fruits).
- Measured variables (BMI, Age).

Suggested exploratory analysis:

- Table of counts and proportions for the binary variables.
- Summary statistics for BMI and Age by diabetes status.
- Simple plots:
  - Bar plots or proportion plots for binary predictors versus diabetes status.
  - Boxplots of BMI and Age by diabetes status.
- Initial discussion of which variables seem related to higher diabetes risk.

You should also:

- Explain what signs you expect in the logistic model coefficients.
- For example: high blood pressure is likely to increase the odds of diabetes; older age likely increases risk, etc.

3.3 3. Initial Logistic Regression Model

Fit a multiple logistic regression model with all six explanatory variables:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \cdot \text{HighBP}\beta_2 \cdot \text{HighChol}\beta_3 \cdot \text{BMI}\beta_4 \cdot \text{Smoker}\beta_5 \cdot \text{Fruits}\beta_6 \cdot \text{Age}$$

In your report:

- State that you are using `glm(..., family = binomial(link = "logit"))`.
- Report the main results from the model:
  - Coefficients, standard errors, `z-values`, `p-values`.
  - Deviance, null deviance, `AIC`.
  - Comment on whether each variable appears to have a significant effect (using `p-values` or `z-statistics`).
  - Comment on whether the signs of the coefficients match your expectations.

You should also briefly mention why linear regression is not appropriate here (binary response, non-normal errors, etc.), and that logistic regression models the log-odds.

3.4 4. Model Selection and Goodness of Fit

Explain how you assess whether the variables collectively contribute to the model, and how you decide which variables to keep.

Typical steps:

- Use the difference between null deviance and residual deviance to form the G statistic (likelihood ratio test).
- Compare this G statistic to the `chi-squared` distribution to test:
  - ($H_0 : \beta_1 = \cdots = \beta_6 = 0$).
  - (H_1 :) at least one ($\beta_i \neq 0$).
- If appropriate, use `anova(m1, test = "Chisq")` to compare nested models and see which variables significantly improve model fit.
- Use `AIC` to compare models and favour more parsimonious ones (lower `AIC` is better, all else equal).

For individual variables:

- Use the `Wald z-statistic` and its `p-value`.

- A rule of thumb: $if(|z| < 2)$ and $p - value > 0.05$, the variable may be a candidate for removal.
- However, you should apply judgement and consider theory as well, not just automatic rules.

Describe clearly which variables you remove, in what order, and why.

3.5 5. Final Logistic Model and Interpretation

Present your chosen final model, for example:

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{HighBP}\hat{\beta}_2 \cdot \text{BMI}\hat{\beta}_3 \cdot \text{Age}$$

Explain:

- The interpretation of coefficients in terms of odds:
  - For a binary variable (e.g. HighBP): how the odds of diabetes change when HighBP = 1 vs 0, holding other variables constant.
  - For a measured variable (e.g. BMI): how the odds change for a one-unit increase in BMI, holding other variables constant.
- If you wish, you can also mention odds ratios (exponentiated coefficients), but keep explanations clear and not overly technical.

3.6 6. Classification Accuracy and Model Performance

Assess the predictive performance of your final logistic model.

Suggested approach:

- Obtain predicted probabilities ($\hat{p}$) for each observation.
- Choose a classification rule (e.g. classify as "diabetes / prediabetes" if

$$(\hat{p} \geq 0.5))$$

  .
- Create a confusion matrix comparing predicted class to actual class.
- Compute:

  ```
  - Overall accuracy (correct predictions / total).
  ```

- Comment on:

  ```
  - Whether the accuracy is high or modest.
  - Any imbalance in prediction errors (e.g. more false negatives than false positives).
  - Any concerns about overfitting (especially if using a very small sample).
  ```

You do not need to show all R code; just summarise the key numeric results and their interpretation.

4 Final Checklist for Your Report

Use the following checklist before submission:

- I have clearly stated the research questions for both Q1 and Q2.

- I have described the data and variables (types, roles, basic summaries).
- I have used appropriate plots to explore relationships and commented on them.
- For Q1, I have:
  - Specified an initial multiple regression model.
  - Explained how categorical variables are encoded (dummy variables).
  - Used `F-tests`, (R^2), and adjusted (R^2) in model evaluation.
  - Discussed multicollinearity where relevant.
  - Reported and interpreted a final model.
  - Produced and commented on the required prediction.
- For Q2, I have:
  - Explained why logistic regression is appropriate.
  - Fitted an initial logistic model with all explanatory variables.
  - Used deviance, `G tests`, `Wald z-statistics`, or `AIC` in model comparison.
  - Reported and interpreted a final logistic model.
  - Assessed classification accuracy (confusion matrix, accuracy rate).
- My report is clear, concise, and within the 2,000-word limit.
- I have explained my reasoning at each step, not just quoted `R` output.

Remember:
The markers are looking not only for correct models, but for evidence that you understand the modelling process and can communicate it clearly.

Good luck!